Can Next Week's Dengue Surge Be Flagged Early? An Early-Warning Model for Singapore

Project 1.1 (public health, intermediate) · Statistical Learning group project · [Group members]

1. Question and motivation

Dengue is endemic in Singapore and arrives in irregular epidemic waves (e.g. 2013–14, 2019–20, 2022). **Primary question:** can recent dengue counts and lagged weather (rainfall, temperature, humidity) identify weeks with a materially elevated risk of a surge in the following week(s)? An early, calibrated risk signal can support inspection, communication and vector-control planning, even though the model does not establish what causes a surge. We frame the task as *prediction*, not causal inference.

2. Data and curation

- **Cases:** MOH weekly dengue fever (DF) and dengue haemorrhagic fever (DHF), data.gov.sg, 2014–2018 (main analysis); MOH Weekly Infectious Disease Bulletin 2012–2022 for an extended robustness analysis covering more outbreaks. DF is the target series; DHF is reported separately (<1 case/week on average).
- **Weather:** NEA/MSS daily records (Changi), aggregated to epidemiological weeks and lagged 0–12 weeks; reanalysis (Open-Meteo) only to fill gaps, cross-checked against NEA monthly series.
- **Curation:** a complete epi-week calendar (Sun–Sat, MMWR numbering, incl. 53-week years); datasets joined on overlapping weeks only; every predictor uses information available at forecast week t.

3. Surge definitions (fixed before any model is fitted)

For horizon $h \in \{1, 2, 4\}$ weeks ($h = 1$ is primary) and DF counts y_t:

- **Seasonal percentile (S):** $y_{t+h} > Q_{0.90}$ of counts in the same season (± 2.5 weeks) of earlier years.
- **Historical threshold (A):** $y_{t+h} > \bar{y}_{t-51:t} + 2\, s_{t-51:t}$ (past 52 weeks).
- **Week-on-week rise (B):** $y_{t+h} \geq (1 + p_h)\, y_t$ and $y_{t+h} \geq 50$.
- **Onset variants of A and S:** only weeks where y_t is *not yet* above the threshold, so that the model is judged on genuine early warnings rather than on the persistence of an ongoing epidemic.

Thresholds are tuned once so that 10–20% of weeks are surges (preliminary: A 19%, S 19%, B 12–18%).

4. Methods

- **Baselines:** base rate; *seasonal* (historical surge rate for that week of the year); *persistence* (current distance to the threshold, or last week's growth).
- **Classification:** logistic regression with lagged cases and/or weather; ℓ_1-regularised logistic regression (penalty by forward-chaining CV) for variable selection; CART (depth 3, interpretable rules); random forest.
- **Count regression:** Poisson GLM and ridge regression on $\log(1 + y)$, converted to $P(\text{surge})$ through a predictive distribution so that all models are compared on the same scale.

5. Validation and evaluation

Rolling-origin evaluation *by week* with an expanding window: at each forecast week the model is trained only on weeks whose outcome is already known (purged by h), with no future weather or later case revisions in the predictors. We report **PR-AUC** (main metric, surges are rare), recall, precision and alarms per year at an alarm threshold set by the cost ratio of a missed surge to a false alarm, **calibration** (Brier score, reliability diagrams), 95% block-bootstrap intervals, and the gain over the persistence and seasonal baselines.

6. Anticipated limitations (defence focus)

- **Reporting and revisions:** only final counts are public; sensitivity run assuming a 1-week reporting delay.
- **Immunity and serotype shifts** (e.g. DENV-3 in 2020) change the case–weather relation; performance is reported per test year.
- **Few large outbreaks:** small effective sample; simple, regularised models and bootstrap intervals.
- **Cost of false alarms:** alarm rules compared across cost ratios (1:1 to 9:1), expressed as alarms per year.

7. Deliverables and timeline

Wk 1–2: data curation and surge definitions · Wk 3–4: baselines, logistic and count models · Wk 5: trees, regularisation, model selection · Wk 6: robustness and defence analyses · Wk 7–8: report and reproducible code (public GitHub pipeline).

References. Hii Y.L. et al. (2012), Forecast of dengue incidence using temperature and rainfall, *PLoS NTD* 6(11):e1908. Shi Y. et al. (2016), Three-month real-time dengue forecast models: an early warning system for outbreak alerts and policy decision support in Singapore, *Environ. Health Perspect.* 124(9):1369–1375.